MUVE QR

SYSTEM CAPABILITIES & SPECIFICATION OVERVIEW

Smart Security Patrol & Facility Guard Tour Management System

FOR MANAGEMENT & OPERATIONS REVIEW

1. Executive Summary

MUVE QR is an enterprise-grade digital guard tour tracking, facility inspection, and emergency dispatch management platform. It replaces outdated paper logbooks and legacy RFID baton hardware with an intelligent, cloud-connected mobile web solution that guarantees real-time guard accountability, incident documentation, and rapid emergency response.

2. Security Guard Mobile Application (Field Operations)

A. Next Checkpoint Guidance (Hero Tour Navigator)

- Giant Pending Checkpoint Card: Displays the exact next checkpoint to patrol (e.g., 01st Floor - Ladies Washroom), eliminating missed checkpoints.
- Tour Progress Indicator: Live visual gauge (e.g., 8 / 10 Points Completed) showing real-time patrol tour completion percentage.
- One-Tap Scan Trigger: Prominently placed on the mobile home screen for quick scanning without navigating menus.

B. High-Speed QR Checkpoint Scanner

- Built-in Flashlight & Camera Scanner: Rapid QR scanning in dark basements, stairwells, and night patrols.
- Haptic Vibration Feedback: Tactile confirmation on successful scans.
- GPS Geolocation Tagging: Captures exact device coordinates per scan to prevent off-site check-in fraud.
- Anti-Spoof Token Verification: Cryptographically secured QR tokens prevent reproduction from photographs.

C. Tactical Emergency SOS & Panic Dispatch System

- 5-Second Safety Hold Activation: Requires holding the SOS button for 5 seconds with a live circular progress countdown to eliminate accidental pocket presses.
- Dual-Tone Tactical Siren Engine: Multi-oscillator emergency alarm (540Hz - 1180Hz rapid frequency sweep + dissonant alarm + sub-bass pulses) engineered via Web Audio API without requiring audio downloads.
- Hardware Vibration Engine: Aggressive pulsing vibration alerts nearby personnel in stealth or high-noise situations.
- Live GPS Emergency Broadcast: Coordinates are immediately dispatched to Headquarters and the Admin Operations Center.
- Full-Screen Strobe Alert & Hotlines: Flashing emergency overlay with one-tap calling to 119 (Police) and 1990 (Ambulance).

D. Mobile Incident & Hazard Reporting

- Live Camera Evidence Attachment: Guards take photos of broken locks, water leaks, or safety hazards directly from their device.
- Auto Image Compression: Client-side optimization ensures instant upload even on low-speed 3G/4G connections.
- Categorization & Severity: Tagged as Security, Fire Safety, Maintenance, Medical, or Hazard with Low/Medium/High/Critical ratings.
- Guard Incident History: Guards can review previously submitted reports and resolution statuses.

E. Task Checklist & PWA Support

- Daily Routine Checklist: Real-time digital tasks (e.g., verify fire extinguishers, test emergency exit doors).
- Progressive Web App (PWA): Installable on Android and iOS devices via "Add to Home Screen" without App Store delays.
- Offline Service Worker Support: Offline caching ensures continuous UI and emergency operation during signal loss.

3. Admin Command Center & Headquarters Dashboard

A. Real-Time Operations & Live Feed

- Executive KPI Counters: Live counts of Total Scans Today, Active Guards on Duty, Pending Checkpoints, and Open Incidents.
- Live Scans Stream: Real-time incoming feed showing guard name, checkpoint location, exact timestamp, and GPS coordinates.

B. Incident Operations Center (/admin/incidents)

- Audible Alarm in Admin Room: Plays a siren on the supervisor computer immediately upon an SOS trigger or critical incident.
- Evidence Photo Viewer: Full-resolution modal for reviewing photos uploaded by field guards.
- Status & Action Logging: Full incident lifecycle management (Open -> In Progress -> Resolved) with supervisor resolution notes.
- Audit CSV Export: One-click export of incident reports for security audits, insurance claims, and client compliance.

C. QR Code & Checkpoint Management

- Multi-Zone Hierarchy: Organize checkpoints by Building, Floor, Zone, and Description.
- Printable Batch QR Sheets: Generates crisp, branded QR codes ready for printing and physical mounting.
- Dynamic Token Rotation: Regulate and rotate checkpoint security tokens at any time.

D. Guard Performance & Accountability Analytics

- Tour Completion Tracking: Pinpoints missed checkpoints, patrol lapses, and incomplete tours.
- Patrol Punctuality & Duration: Tracks time elapsed between checkpoints to prevent rushing or patrol skipping.

E. User Management & Access Control (RBAC)

- Role Permissions: Strict separation between Administrators (HQ control, reports) and Officers/Guards (mobile patrol only).
- Audit Trail: Immutable timestamped log of all patrol activities, logins, and system changes.
- Management Reports: Generate daily, weekly, and monthly summaries with CSV/PDF export.

4. Key Business Benefits for the Organization

Key Benefit	How MUVE QR Delivers
100% Guard Accountability	Eliminates falsified paper logbooks. Every checkpoint scan is cryptographically signed with GPS coordinates and tamper-resistant timestamps.
Zero Dedicated Hardware Cost	Runs on guards' existing Android or iOS smartphones. No expensive proprietary batons, RFID wands, or docking stations required.
Officer Life Safety & Panic Response	5-second hold SOS button with loud dual-tone emergency siren and instant GPS broadcast ensures rapid backup during attacks or medical emergencies.
Real-Time Facility Maintenance	Hazards, leaks, and damages are photographed and flagged immediately, minimizing facility downtime and property liability.
Client Transparency & SLA Proof	Instantly generate verified PDF and CSV reports to prove contractual security SLA compliance to property owners and auditors.